

Seeing Monitor Using Stellar Image Motion

We have a small lens with diameter D_m and focal length F_m , and a camera with pixel size Pix . All values are in meters. The image scale of the system is

$$\text{Scale} = \frac{206265 Pix}{F_m} \quad (\text{arcsec/pixel}). \quad (1)$$

Using this system, we obtain a statistically significant number of images (100–1000 or more) of Polaris, without guiding, using short exposures. Care should be taken to avoid saturation.

For each frame, we determine the precise (X, Y) position of the center of the stellar image using either a center-of-mass calculation (centroiding) or profile fitting.

Because the star moves from one exposure to the next, this trend must be removed. For short sequences (approximately 1 minute), fitting X and Y with a second-order polynomial is sufficient. In this case, the residual error will be sufficiently small (Fig. 1).

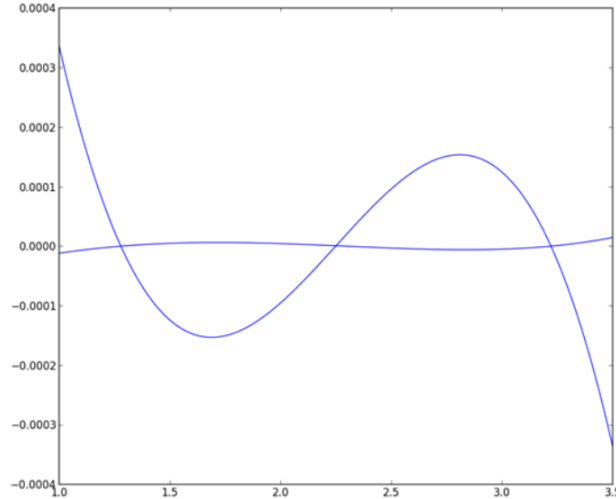


Figure 1: Residual error (pixels) after fitting a second-order polynomial to a 60-second sequence of frames; Scale = 5.5 arcsec/pixel.

After subtracting the corresponding trends from X and Y , we assume that the data contain only random motion of the star in the focal plane caused by atmospheric wavefront tilts (Fig. 2).

The quantity characterizing the scatter of the points is the variance, σ^2 . The variance is calculated as

$$\sigma^2 = \text{Var} = \frac{1}{n} \sum_{k=0}^n (X_k - \langle X \rangle)^2. \quad (2)$$

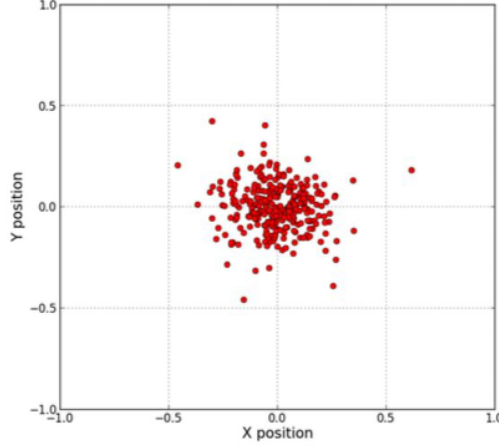


Figure 2: Scatter of the stellar-center coordinates caused by atmospheric image motion; $D_m = 0.05$ m, Scale = 5.5 arcsec/pixel.

In our case, after removal of the trend, the mean coordinate value is zero. The square root of the variance is the standard deviation.

We are interested in the variance of the stellar-center displacement in each frame. It can be calculated as the sum of the variances in X and Y :

$$\sigma_*^2 = \sigma_X^2 + \sigma_Y^2 = \frac{1}{n} \sum_{k=0}^n (X_k^2 + Y_k^2). \quad (3)$$

References [1, 2, 3] provide equations relating the variance of the image-centroid position, the telescope diameter D_m , and the Fried parameter r_0 :

$$\sigma_*^2 = K\lambda^2 \left(\frac{1}{r_0}\right)^{5/3} \left(\frac{1}{D_m}\right)^{1/3}. \quad (4)$$

An important conclusion from Eq. (4) is that a **small-aperture lens is preferable for constructing a seeing monitor**. In this case, the variance increases and can be measured more accurately.

All subsequent calculations are performed for

$$\lambda = 550 \text{ nm} = 5.5 \times 10^{-7} \text{ m}.$$

The parameter K depends on the ratio of the separation between the apertures to the aperture diameter (for a DIMM). For zero separation—that is, for a single aperture in our case—

$$K = 0.358.$$

The Fried parameter is then calculated as

$$r_0 = \left(\frac{0.358\lambda^2}{\sigma_*^2}\right)^{0.6} \left(\frac{1}{D_m}\right)^{0.2}. \quad (5)$$

The Fried parameter determines the *seeing*, i.e. the FWHM of an image limited by atmospheric turbulence for a telescope of arbitrary diameter:

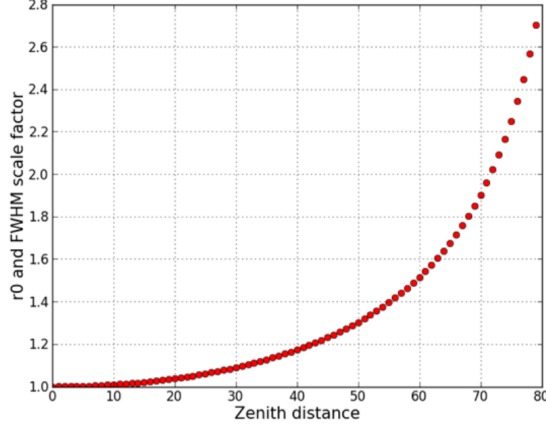


Figure 3: Variation of the Fried parameter and FWHM with zenith distance.

$$\text{FWHM}(\text{rad}) = 0.98 \frac{\lambda}{r_0}, \quad (1)$$

or

$$\text{FWHM}(\text{arcsec}) = 0.98 \frac{\lambda}{r_0} \times 206265. \quad (6)$$

This relation is very similar to the definition of the Rayleigh criterion (angular resolution), except that the Fried parameter is used instead of the telescope aperture diameter.

The values obtained from Eqs. (5) and (6) correspond to the zenith distance Z of the observed star—in our case, Polaris.

To obtain values corresponding to the zenith, the change in airmass with zenith distance must be taken into account. The Fried parameter and seeing scale with airmass to the power of 0.6:

$$\text{FWHM}_z = \frac{\text{FWHM}_p}{(\cos Z)^{-0.6}}, \quad (2)$$

and

$$r_{0z} = r_{0p}(\cos Z)^{-0.6}. \quad (7)$$

The dependence is shown in Fig. 3. At an altitude of 20 degrees above the horizon, the image quality is approximately twice as poor as at the zenith.

References

- [1] M. Sarazin and F. Roddier, “The ESO differential image motion monitor,” *Astronomy and Astrophysics*, vol. 227, no. 1, pp. 294–300, Jan. 1990.
- [2] A. Tokovinin, “From Differential Image Motion to Seeing,” *Publications of the Astronomical Society of the Pacific*, vol. 114, pp. 1156–1166, Oct. 2002.
- [3] M. Sarazin, “Measuring Seeing, The Differential Image Motion Monitor (DIMM),” ESO presentation.
<https://www.eso.org/astclim/espas/iran/zanjan/zanjan02.ppt>